



Quality Control (QC) Validating Report

Software Requirements Specification (SRS)

Biometric Student Attendance System

Veridion Labs

March 20, 2026

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Table 1: Veridion Labs QC Audit Team

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1 Executive Summary

This report presents the results of a Quality Control (QC) validation of the Software Requirements Specification (SRS) for the **Biometric Student Attendance System**, prepared by Nevon Solutions for BioSec Educational Institution (NS-SRS-2026-001, Version 0.8, 3 March 2026). The audit focused on three principal categories of SRS defects expected at the Validating stage: **authorship** (unclear expression), **omission** (missing content), and **contradiction** (inconsistent behaviour across sections).

The SRS is generally well-structured and covers most core functional areas. However, defects were detected including contradictions in 2FA scope and attendance status terminology, omissions for biometric enrolment, password recovery, offline limit handling, and session scheduling, as well as grammatical and editorial errors. The SRS sign-off section also contains blank approver signatures.

These findings indicate that the SRS is **conditionally acceptable**, subject to BioSec's review. No recommendations for correction are provided; the customer will determine any necessary actions.

2 Introduction

2.1 Purpose

This Quality Control (QC) Validating Report presents the findings of an independent inspection of the Software Requirements Specification (SRS) prepared for the **Biometric Student Attendance System** (NS-SRS-2026-001). The purpose of this QC activity is to identify defects remaining in the SRS work product before the project progresses into downstream stages such as system design and implementation.

In accordance with the validating stage of Requirements Engineering, QC focuses on defect detection in the **requirements work product itself**. The QC team records findings but does not propose fixes; the customer determines any corrective actions.

2.2 Document Conventions and Intended Audience

This document is a validating report rather than a replacement SRS. It records detected defects and explains their implications. Audit findings use the format QC-FND-XXX and error detection instances use the format QC-ERR-XXX. This report is intended for the BioSec requirements engineering team, project management personnel, and relevant stakeholders. Readers should begin with the Executive Summary and Sections 1–3, then proceed to the QC Checklist, Error Detection, and detailed Findings.

2.3 Audit Scope

The audit covers the SRS (NS-SRS-2026-001) as a QC exercise. The scope includes:

- detection of authorship defects (ambiguity, vagueness, grammatical errors)
- detection of latent contradictions across sections
- detection of omission of topics (missing scenarios, exception paths, uncovered capabilities)
- checking codification and traceability coherence
- checking structural, editorial, and presentation consistency

3 Audit Basis

3.1 Basis, Error Types, and Severity

This audit assumes the SRS is mature after earlier RE stages. QC inspects the work product itself, identifying defects without direct amendment. The audit team reports findings for customer consideration. The following error types were pursued:

- **Authorship:** unclear expression of intended meaning.
- **Latent contradictions:** conflicting requirements across sections.
- **Omission:** missing topics, edge conditions, or supporting information.
- **Traceability:** incomplete codification, source mapping, or cross-references.
- **Editorial:** formatting, naming, or presentation inconsistencies.

Each finding is classified by severity: **High** (significantly affects interpretation, implementation, or expected behaviour), **Medium** (introduces ambiguity without changing core intent), or **Low** (minor editorial issues with no material impact).

3.2 Documents Reviewed

- SRS: Biometric Student Attendance System (NS-SRS-2026-001, Version 0.8, 3 March 2026)
- BioSec Project Specification (BS-PS-2026-001, Version 1.0)
- Requirements models: Context DFD, Wireframes, Use Case Diagram, ERD (Figures 1–4)
- Inceptive Requirements Traceability Matrix (iRTM, Appendix D)
- Change Mitigation Strategies (Section 8)
- Data Dictionary (Appendix C) and Glossary (Appendix A)

4 QC Plan

4.1 Target Workproduct, Context, and Access

The target workproduct is the **Software Requirements Specification (SRS)** for the Biometric Student Attendance System, prepared by Nevon Solutions for BioSec Educational Institution (NS-SRS-2026-001, Version 0.8). The SRS serves as the contractual baseline for all downstream design, development, testing, and delivery activities. This audit verifies the SRS's clarity, completeness, and internal consistency, documenting any detected defects for BioSec's consideration and action. The QC team reviewed the released SRS document and all supporting artefacts provided for audit, including diagrams, appendices, the iRTM, and change mitigation strategies.

4.2 Inspection Techniques Used

The QC audit employed investigative techniques organised by the error type each technique targets: authorship detection, omissions detection, and contradictions detection. The full mapping of techniques to error types is presented in [Appendix A, Table A-3: Inspection Techniques Used](#).

4.3 Work Breakdown, Timeline, and Reporting

Team roles and responsibilities are detailed in [Appendix A, Table A-1: QC Audit Team Work Allocation](#). The audit schedule is shown in [Appendix A, Table A-2: Indicative QC Audit Timeline](#). The QC team records findings in this report; the audit team does not amend the SRS.

5 QC Checklist and Error Detection

The QC audit team applied a tailored inspection checklist of 21 items derived from industry standards and lecture guidelines, covering correctness, completeness, consistency, relevance, measurability, and traceability across all SRS sections. The full checklist register is presented in [Appendix B, Table B-1: QC Inspection Checklist](#).

The audit identified instances of three principal QC error types: authorship, omission, and contradiction. Each detected instance was recorded with its location and defect characteristic. The error detection register is presented in [Appendix C, Table C-1: Detected QC Error Instances](#), and a summary of all findings in [Appendix D, Table D-1: QC Findings Summary](#).

6 QC Findings

6.1 Finding Summary

The audit identified 12 findings across three defect types. A consolidated summary of all findings is provided in [Appendix D, Table D-1: QC Findings Summary](#). Detailed analysis for each finding is presented in Section [6.2](#) below.

6.2 Detailed Findings

QC-FND-001: Contradiction in Two-Factor Authentication Scope

Two conflicting requirements exist on which user classes must use 2FA. UR-MOB-06 mandates 2FA for parents and teachers on the mobile app, while NFR-SEC-03 restricts 2FA only to administrators on the web portal. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-1a and E-1b](#).

Related error detection: QC-ERR-001

Type: Contradiction

Severity: High

Context. Section 3.1 UR-MOB-06 requires a 2-Factor Authentication screen process for “parent and teacher” mobile application users. Section 5.3 NFR-SEC-03 separately mandates that the system “shall enforce two-factor authentication (2FA) for all administrator accounts accessing the web portal” – with no mention of parents or teachers.

Result. A developer cannot determine whether 2FA is required for all three user classes or only administrators. Implementing 2FA for parents and teachers incurs additional development effort and user experience considerations not reflected in the NFR.

Analysis. These two requirements are mutually inconsistent on the scope of 2FA enforcement. One must be incorrect or the SRS must be clarified to specify a single authoritative position on which user classes require 2FA.

QC-FND-002: Contradiction in Attendance Status Terminology

The attendance status value “Attended” used in the UI requirements conflicts with “Present” used in the AttendanceRecord data dictionary ENUM, creating a direct inconsistency between the presentation layer and the data layer. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-2a to E-2d](#).

Related error detection: QC-ERR-002

Type: Contradiction

Severity: High

Context. Section 3.1 (UR-MOB-01) defines attendance status values as “Attended, Absent, Late, or Excused”. Section 4.3.2 (Stimulus/Response Sequences) repeats “Excused, Attended, Late, Absent”. However, Appendix C (AttendanceRecord table) defines the status ENUM as “Present, Late, Absent, Excused”.

Result. The system defines two different terms (“Attended” and “Present”) for the same attendance state across different parts of the SRS, resulting in inconsistency between the presentation layer and the data layer.

Analysis. Developers implementing the database schema will adopt “Present” as defined in the data dictionary, while UI and functional requirements specify “Attended”. This mismatch will lead to inconsistent data representation, mapping issues, and potential defects during system integration and testing.

QC-FND-003: Contradiction in Report Generation Time Specification

The response time for report generation is described inconsistently between functional behaviour and non-functional requirements. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-3a and E-3b](#).

Related error detection: QC-ERR-003

Type: Contradiction

Severity: Medium

Context. Section 4.6.2 (Stimulus/Response Sequences) states that the system generates attendance reports within “10–60 seconds depending on data volume”. However, NFR-PERF-03 specifies two distinct performance thresholds: 10 seconds for single-class queries and 60 seconds for institution-wide reports.

Result. The functional description presents a generalised timing range without distinguishing between different query types, while the non-functional requirement defines specific measurable conditions.

Analysis. This inconsistency creates ambiguity for developers and testers. It is unclear which timing constraint applies under specific conditions, leading to potential misinterpretation during implementation or incorrect validation during testing.

QC-FND-004: Omission of Biometric Enrolment Process

The SRS specifies biometric attendance functionality but does not define how users are initially enrolled into the biometric system. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-4a and E-4b](#).

Related error detection: QC-ERR-004

Type: Omission

Severity: High

Context. The SRS describes fingerprint-based attendance recording across Sections 3.2, 4, and Appendix C. However, there is no requirement specifying how students are initially registered in the system, including fingerprint capture, responsible actors, enrolment conditions, or error handling.

Result. A prerequisite capability required for biometric attendance operation is missing. Without enrolment, students cannot be recognised by the system, making attendance recording infeasible.

Analysis. This is a critical omission defect. Biometric enrolment is a foundational process required before any attendance functionality can operate. Its absence forces developers to make assumptions about system behaviour, leading to inconsistent implementations and increased risk of system failure.

QC-FND-005: Omission of Password Reset and Account Recovery

The SRS specifies account lockout after failed login attempts but provides no requirement for how locked-out users can recover access, leaving a critical gap in the authentication lifecycle. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-5a and E-5b](#).

Related error detection: QC-ERR-005

Type: Omission

Severity: High

Context. NFR-SEC-04 specifies that the system shall lock user accounts after a configurable number of failed login attempts. Section 4.2.2 Stimulus/Response confirms that after five unsuccessful attempts, the account is locked. However, no requirement in the SRS defines how a user recovers access – there is no password reset, account unlock, or self-service recovery flow specified anywhere.

Result. Users who trigger the lockout policy have no defined path to regain access. Administrators, parents, and teachers could be permanently locked out with no recovery mechanism, directly impacting system usability and operational continuity.

Analysis. Account lockout without a corresponding recovery mechanism is an incomplete functional capability. This omission will require resolution before the system can be deployed to real users.

QC-FND-006: Omission of Offline Limit Exceeded Handling

The SRS defines a 48-hour offline mode but contains no requirement for system behaviour when this limit is exceeded or for resolving data conflicts upon reconnection. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-6a and E-6b](#).

Related error detection: QC-ERR-006

Type: Omission

Severity: High

Context. Section 2.5 specifies that the system must support offline attendance viewing and leave request drafting for up to 48 hours, with a 50MB local storage limit. NFR-SEC-09 confirms cached data is auto-purged after 48 hours. However, there is no requirement specifying what happens when the 48-hour offline limit is exceeded mid-session, or how data synchronisation conflicts are resolved when connectivity is restored after an offline period.

Result. If a user exceeds the 48-hour offline limit, the system behaviour is undefined. Similarly, if locally drafted leave requests conflict with server-side data upon reconnection, there is no specified resolution strategy.

Analysis. Offline mode is an explicitly scoped capability. The absence of requirements for limit-exceeded behaviour and conflict resolution leaves developers to make assumptions that may be inconsistent with stakeholder expectations.

QC-FND-007: Omission of Session Creation and Management Requirements

Academic sessions are referenced extensively as a prerequisite for attendance capture, but no requirement in the SRS defines how sessions are created, scheduled, or managed within the system. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-7a and E-7b](#).

Related error detection: QC-ERR-007

Type: Omission

Severity: Medium

Context. BR-02 states that attendance capture is only permitted during “officially scheduled academic sessions” and that scans outside session windows are rejected. FR-ATT-03 references a “session_id” as a stored field in AttendanceRecord. The AttendanceRecord data dictionary (Appendix C, Table C.4) also references a “session_id FK to scheduled academic session”. However, no requirement anywhere in the SRS defines how academic sessions are created, configured, scheduled, or managed within the system.

Result. The system cannot enforce BR-02 or populate session_id without sessions being defined in the system. The SRS assumes sessions exist but never specifies how they come to exist.

Analysis. This is a functional gap. Session management is a prerequisite for the attendance capture business rule to function correctly.

QC-FND-008: Ambiguous Use of “Periodic Synchronisation”

The product functions summary uses the vague term “periodic synchronisation” without specifying a frequency, inconsistent with the precise 5-minute figure stated in FR-COM-01. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figures E-8a and E-8b](#).

Related error detection: QC-ERR-008

Type: Authorship

Severity: Medium

Context. Section 2.2 Product Functions describes attendance data access as providing “role-scoped mobile access for parents (child only) and teachers (assigned classes only), with periodic synchronisation”. FR-COM-01 specifies that mobile applications shall synchronise attendance data with the backend “every 5 minutes when connected”.

Result. The word “periodic” in Section 2.2 is vague and does not communicate the 5-minute interval. A reader relying on the product functions summary alone would not know the synchronisation frequency.

Analysis. While FR-COM-01 is precise, the inconsistency between vague language in the product summary and specific language in the functional requirement reduces overall document clarity.

QC-FND-009: Grammatical Error in QA-USAB-01

A double verb construction in QA-USAB-01 – “shall allow ... must be able to” – is grammatically incorrect and introduces ambiguity about the nature of the obligation. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figure E-9](#).

Related error detection: QC-ERR-009

Type: Authorship

Severity: Low

Context. Section 5.4.1 QA-USAB-01 states: “The system shall allow parent and teacher users must be able to complete primary tasks ... within three interactions from the home screen”.

Result. The double verb construction “shall allow ... must be able to” is grammatically incorrect and creates ambiguity about the nature of the obligation – whether the system is the subject of the requirement or the users are.

Analysis. While the intended meaning is reasonably clear from context, the grammatical error introduces imprecision in a quality attribute requirement that should be unambiguous for verification purposes.

QC-FND-010: Capitalisation Inconsistency in UR-WEB-08 and UR-WEB-09

Requirements UR-WEB-08 and UR-WEB-09 begin with a lowercase “the portal shall” while all other requirements in Section 3.1 consistently use the capitalised “The portal shall”, indicating an editorial oversight. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figure E-10](#).

Related error detection: QC-ERR-010

Type: Authorship

Severity: Low

Context. Section 3.1 requirements UR-WEB-08 and UR-WEB-09 begin with lowercase: “the portal shall display an activity log...” and “the portal shall include a 2-Factor Authentication screen...”. All other interface requirements in the same section (UR-GUI-01 through UR-WEB-07 and UR-WEB-10) begin with a capitalised “The”.

Result. The inconsistency reduces document professionalism and suggests an editorial oversight in the final review of Section 3.1.

Analysis. While this does not affect the meaning of the requirements, it is indicative of an insufficient final editorial pass prior to SRS release.

QC-FND-011: Unsigned SRS Sign-off Section

All four designated approvers in the SRS sign-off section – two from Nevon Solutions and two from BioSec – have entirely blank signature and date fields, indicating the SRS was released without their formal approval. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figure E-11](#).

Related error detection: QC-ERR-011

Type: Authorship

Severity: Low

Context. Section 9 SRS Sign-off contains a Designated Approvers table for both Vendor (Nevon Solutions: James Koh – Technical Lead, Rachel Lee – System Architect Lead) and Customer (BioSec: Marcus Tan Wei Liang – Project Manager, Daniel Rodriguez – Head of Operations). All four approver rows have entirely blank Signature and Date fields.

Result. The SRS was released without designated approver sign-off. The document’s own sign-off certifies that it “constitutes the approved development baseline”, but the approver table contradicts this by showing no approvals.

Analysis. While the Project Lead (Loh Wen Xuan) has signed the Vendor Authenticator section, the absence of designated approver signatures raises a question about whether the SRS was formally reviewed and approved by all required parties before release.

QC-FND-012: RBAC Not Defined in Glossary

RBAC is one of the most frequently referenced acronyms in the SRS but is absent from Appendix A Glossary, which defines 27 other terms including abbreviations of similar nature. See [Appendix D, Table D-1: QC Findings Summary](#) for the finding summary entry. Evidence: [Appendix E, Figure E-12](#).

Related error detection: QC-ERR-012

Type: Omission

Severity: Medium

Context. The acronym RBAC (Role-Based Access Control) is used throughout the SRS – in Section 4.1 heading, FR-RBAC-01, QA-INTEG-01, and multiple other locations. Appendix A Glossary defines

27 terms including FR, NFR, BOR, TR, and other abbreviations, but RBAC is not among them.

Result. A reader unfamiliar with RBAC would not find a definition in the SRS itself despite RBAC being central to the system's security architecture.

Analysis. Given that RBAC is one of the most frequently referenced concepts in the SRS and the Glossary explicitly defines other abbreviations of similar nature, the omission of RBAC is an inconsistency in the addenda.

7 Conclusion

The SRS is substantively well-structured with thorough coverage of core functional areas. However, the audit identified 12 findings across three defect types, including high-severity contradictions in 2FA scope and attendance status terminology, and high-severity omissions for biometric enrolment, password recovery, and offline limit handling. The findings are presented for BioSec's consideration; the QC team does not amend the SRS.

Overall, the SRS is assessed as **conditionally acceptable**, pending customer review and resolution of the identified defects.

8 Sign-off

8.1 QC Audit Team Authentication

Name	Role	Authentication
Muhammad Ridwan Putra Jasni	QC Audit Lead	
Lin Yuhao	QC Auditor	
Muhammad Mikhail Bin Mazlan	QC Auditor	
Kannan s/o Rajamohan	QC Auditor	

Table 2: QC Audit Team Sign-Off

9 Addenda

9.1 Glossary

Term	Definition
QC	Quality Control – independent inspection of a work product to detect defects, in this case the SRS.
SRS	Software Requirements Specification – the requirements baseline document for downstream development.
Authorship defect	Defect caused by unclear, imprecise, or grammatically incorrect expression of an intended requirement meaning.
Omission defect	Defect in which required content, scenario, capability, or supporting information is absent from the SRS.
Contradiction defect	Defect in which different parts of the SRS imply inconsistent or conflicting system behaviour.
Traceability coherence	The degree to which requirements, models, sources, and cross-references link together clearly and consistently throughout the SRS.
2FA	Two-Factor Authentication – a security mechanism requiring two forms of verification before granting access.
RBAC	Role-Based Access Control – a method of restricting system access based on the roles assigned to users.
PDPA	Personal Data Protection Act 2012 (Singapore) – the legislation governing collection, use, and disclosure of personal data.
iRTM	Inceptive Requirements Traceability Matrix – a matrix tracing PS requirements to SRS requirements.

Table G-1: QC Report Glossary

Appendix A: QC Plan Tables

A.1 QC Audit Team Work Allocation

Role	Assigned Member	Primary Responsibility
QC Audit Lead	Muhammad Ridwan Putra Jasni	Coordinate audit activities, consolidate findings, ensure consistency of final report, apply cross-section contradiction checking and assumption reading.
Requirements Reviewer	Lin Yuhao	Inspect functional and non-functional requirement sections for ambiguity, contradiction, and omission; perform perspective-based reading (developer, tester) and scenario-based reading for biometric attendance and leave management flows.
Traceability Reviewer	Muhammad Mikhail Bin Mazlan	Review iRTM references, source mappings, data dictionary, and section-to-section coherence; double-check all identifiers, cross-references, and figure citations.
Editorial and Structure Reviewer	Kannan s/o Rajamohan	Review document structure, section naming, formatting consistency, and overall presentation quality; conduct spellcheck and grammar review, verify figures, tables, and sign-off completeness.

Table A-1: QC Audit Team Work Allocation

A.2 Indicative QC Audit Timeline

Stage	Activity	Indicative Duration
Stage 1	Individual inspection of assigned SRS sections using selected techniques (close reading, perspective-based, scenario-based)	3 hours
Stage 2	Consolidation meeting and de-duplication of identified findings	1 hour
Stage 3	Severity review and documentation of findings	1 hour
Stage 4	Final report formatting, checking, and team sign-off	1 hour

Table A-2: Indicative QC Audit Timeline

A.3 Inspection Techniques Used

Technique	Error Type Targeted
Close document inspection	Authorship Detection
Spellcheck and grammar review	
Double-checking dates, facts, figures, and identifiers	
Skimming (top-down & bottom-up)	Omissions Detection
Assumption reading	
Scenario-based reading	
Perspective-based reading	
Cross-section contradiction checking	Contradictions Detection
Persona-based reading	
Codification and traceability verification	

Table A-3: Inspection Techniques Used

Appendix B: QC Checklist

Item	Focus	QC Checklist Item	Inspection Result
1. Introduction Section			
QC1	Correctness	Does the Introduction explicitly define the system purpose, target user base, and operational boundaries (in-scope vs. out-of-scope)?	Satisfied
QC2	Completeness	Does the Introduction identify all intended audiences and provide reading guidance for each?	Satisfied
QC3	Consistency	Is the system name, document ID, version number, and project title consistent across the title page, header, and body?	Satisfied
2. Glossary / Definitions Section			
QC4	Completeness	Are all domain-specific acronyms and abbreviations used in the SRS defined in the Glossary?	Partially satisfied (RBAC missing)
QC5	Relevance	Does every term defined in the Glossary appear at least once in the SRS body?	Satisfied
QC6	Consistency	Is each system role or entity referred to by a consistent term throughout the document?	Partially satisfied
3. Overall Description Section			
QC7	Completeness	Are all user classes identified with their technical expertise, access privileges, and responsibilities?	Satisfied
QC8	Completeness	Does the SRS specify the intended operational environment (OS, platform, hardware, network)?	Satisfied
QC9	Correctness	Are product function descriptions measurably correct and consistent with detailed requirements in later sections?	Partially satisfied
4. Functional Requirements Section			
QC10	Correctness	Does each functional requirement describe a single, testable behaviour using precise language?	Partially satisfied
QC11	Completeness	Does the SRS include requirements for all critical system processes, including biometric enrolment?	Not satisfied
QC12	Consistency	Is every requirement assigned a unique identifier following a consistent naming convention?	Satisfied
5. Interface Requirements Section			
QC13	Completeness	Does the Interface section describe purpose, data exchanged, and communication method for each external interface?	Satisfied
QC14	Correctness	Are trigger conditions, expected data payloads, and error handling defined for each external interface?	Partially satisfied
QC15	Consistency	Do all interface requirements use consistent capitalisation and grammatical structure?	Not satisfied
6. Non-Functional Requirements Section			
QC16	Measurability	Are all NFRs quantified with measurable, testable limits (e.g., response time, availability percentage)?	Partially satisfied
QC17	Completeness	Does the SRS address all relevant NFR categories (performance, security, safety, reliability, usability, maintainability)?	Satisfied

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Item	Focus	QC Checklist Item	Inspection Result
QC18	Consistency	Are NFR obligations internally consistent – e.g., 2FA scope consistent across UR-MOB-06 and NFR-SEC-03?	Not satisfied
7. Addenda / Supporting Artefacts			
QC19	Traceability	Does the iRTM map all PS requirements to corresponding SRS requirements with correct cardinality?	Satisfied
QC20	Correctness	Does every diagram include a descriptive title, revision note, and standard notation accurately representing the system?	Satisfied
QC21	Completeness	Is the SRS sign-off section fully completed with signatures and dates from all designated approvers?	Not satisfied

Table B-1: QC Inspection Checklist

Appendix C: Error Detection Table

Error ID	Error Type	Location	Characteristic of Error
QC-ERR-001	Contradiction	Section 3.1 UR-MOB-06 vs Section 5.3 NFR-SEC-03	UR-MOB-06 requires a 2-Factor Authentication screen for “parent and teacher” mobile app users. NFR-SEC-03 enforces 2FA only for “all administrator accounts accessing the web portal”. These statements directly conflict on which user classes require 2FA.
QC-ERR-002	Contradiction	Section 3.1 UR-MOB-01 vs Section 4.3.2 vs Appendix C Table C.4	UR-MOB-01 and Section 4.3.2 use the status value “Attended”, while the AttendanceRecord data dictionary (Table C.4) defines the status ENUM as “Present, Late, Absent, Excused” – using “Present” instead of “Attended”. The UI layer and data layer use different values for the same status.
QC-ERR-003	Contradiction	Section 4.6.2 Stimulus/Response Sequences vs NFR-PERF-03	Section 4.6.2 states reports are returned “within 10–60 seconds depending on data volume” without distinguishing between query types. NFR-PERF-03 specifies 10 seconds for single class queries and 60 seconds for institution-wide reports. The omission of this distinction in Section 4.6.2 creates an inconsistency with the NFR.
QC-ERR-004	Omission	Entire SRS – Sections 3.2, 4, Appendix C	The SRS specifies extensive biometric scanning behaviour but contains no requirement for the student biometric enrolment process – how fingerprints are initially registered, who performs the enrolment, what constitutes a successful enrolment, or how enrolment failures are handled.
QC-ERR-005	Omission	Section 4.2, NFR-SEC-04	NFR-SEC-04 specifies account lockout after failed login attempts, and Section 4.2.2 confirms lockout after five failures. However, no requirement defines a password reset, account unlock, or self-service recovery process. Users who trigger lockout have no defined path to regain access.
QC-ERR-006	Omission	Section 2.5, NFR-SEC-09	Section 2.5 and NFR-SEC-09 specify a 48-hour offline mode with auto-purge of cached data. No requirement specifies what happens when this limit is exceeded mid-session, nor how data sync conflicts are resolved when connectivity is restored after an offline period.

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Error ID	Error Type	Location	Characteristic of Error
QC-ERR-007	Omission	BR-02, FR-ATT-03, Appendix C Table C.4	BR-02 states attendance capture is only permitted during “officially scheduled academic sessions”. FR-ATT-03 and the AttendanceRecord data dictionary reference a session_id. However, no requirement defines how academic sessions are created, scheduled, or managed in the system.
QC-ERR-008	Authorship	Section 2.2 Product Functions vs FR-COM-01	Section 2.2 describes attendance data access with “periodic synchronisation” – a vague term with no frequency specified. FR-COM-01 specifies synchronisation every 5 minutes when connected. The imprecise language in Section 2.2 is inconsistent with the specific figure in FR-COM-01.
QC-ERR-009	Authorship	Section 5.4.1 QA-USAB-01	QA-USAB-01 states “The system shall allow parent and teacher users must be able to complete primary tasks . . . within three interactions”. The double verb construction “shall allow . . . must be able to” is grammatically incorrect, creating ambiguity about the nature of the obligation.
QC-ERR-010	Authorship	Section 3.1 UR-WEB-08 and UR-WEB-09	UR-WEB-08 begins “the portal shall display an activity log. . .” and UR-WEB-09 begins “the portal shall include a 2-Factor Authentication screen. . .” – both use lowercase “the”. All other requirements in Section 3.1 (UR-GUI-01 through UR-WEB-07, UR-WEB-10) begin with the capitalised “The”.
QC-ERR-011	Authorship	Section 9 SRS Sign-off	The Designated Approvers table lists four approvers – James Koh (Technical Lead), Rachel Lee (System Architect Lead), Marcus Tan Wei Liang (BioSec Project Manager), and Daniel Rodriguez (BioSec Head of Operations) – all with entirely blank Signature and Date fields, indicating the SRS was released without their formal approval.
QC-ERR-012	Omission	Appendix A Glossary vs Sections 4.1, FR-RBAC-01, QA-INTEG-01	The acronym RBAC is used extensively throughout the SRS including in the Section 4.1 heading, FR-RBAC-01, and QA-INTEG-01. Appendix A Glossary defines 27 terms but does not include RBAC, leaving the acronym undefined within the SRS.

Table C-1: Detected QC Error Instances

Appendix D: QC Findings Summary

Finding ID	Type	Severity	Related Error	Short Description
QC-FND-001	Contradiction	High	QC-ERR-001	2FA required for parents and teachers in UI requirements but only for administrators in security NFRs.
QC-FND-002	Contradiction	High	QC-ERR-002	Attendance status terminology inconsistent – “Attended” vs “Present” used across sections and data dictionary.
QC-FND-003	Contradiction	Medium	QC-ERR-003	Report generation time stated without query-type distinction in stimulus/response, contradicting NFR-PERF-03.
QC-FND-004	Omission	High	QC-ERR-004	No requirement for biometric enrolment process – how students are initially registered in the system.
QC-FND-005	Omission	High	QC-ERR-005	Account lockout specified but no password reset or account recovery flow requirement exists.
QC-FND-006	Omission	High	QC-ERR-006	Offline mode limit specified but no requirement for what happens when limit is exceeded or sync conflicts arise.
QC-FND-007	Omission	Medium	QC-ERR-007	Sessions referenced throughout but no requirement defining how sessions are created or managed.
QC-FND-008	Authorship	Medium	QC-ERR-008	“Periodic synchronisation” in Section 2.2 is vague and inconsistent with the specific 5-minute figure in FR-COM-01.
QC-FND-009	Authorship	Low	QC-ERR-009	Grammatical error in QA-USAB-01 – double verb construction “shall allow . . . must be able to”.
QC-FND-010	Authorship	Low	QC-ERR-010	UR-WEB-08 and UR-WEB-09 use lowercase “the portal shall” inconsistent with all other requirements.
QC-FND-011	Authorship	Low	QC-ERR-011	SRS sign-off section contains entirely blank approver signatures and dates for all four designated approvers.
QC-FND-012	Omission	Medium	QC-ERR-012	RBAC used extensively throughout but not defined in Appendix A Glossary.

Table D-1: QC Findings Summary

Appendix E: SRS Evidence Screenshots

This appendix presents annotated screenshots from the SRS (NS-SRS-2026-001, Version 0.8) as supporting evidence for each QC finding documented in Section [6](#).

E.1 QC-FND-001: Contradiction in Two-Factor Authentication Scope

- **UR-MOB-06:** The parent and teacher authentication shall include a 2-Factor Authentication screen process.

Figure E-1a: SRS Section 3.1 – UR-MOB-06 requires 2FA for parent and teacher mobile users.

- **NFR-SEC-03:** The system shall enforce two-factor authentication (2FA) for all administrator accounts accessing the web portal. (TR-1-BE-03; TR-1-CPR-01)

Figure E-1b: SRS Section 5.3 – NFR-SEC-03 enforces 2FA only for administrator accounts on the web portal.

E.2 QC-FND-002: Contradiction in Attendance Status Terminology

- **UR-MOB-01:** The parent interface shall display monthly attendance records for the parent's registered child only, with attendance status shown as Attended, Absent, Late, or Excused.

Figure E-2a: SRS Section 3.1 – UR-MOB-01 uses status value “Attended”.

4.3.2 Stimulus/Response Sequences

- The authenticated parent or teacher accesses the attendance screen.
- The system retrieves the user's role and displays attendance records limited to the authorised scope based on the child.

Figure E-2b: SRS Section 4.3.2 – Stimulus/Response Sequences repeats “Attended”.

AttendanceRecord					
Field name	Data type	Size	Allow null?	Description	Example
attendance_id	INT	11	No	PK: Unique record ID	500001
student_id	INT	11	No	FK to Student	2026001
teacher_id	INT	11	No	FK to Teacher (who took attendance)	T001
session_id	INT	11	No	FK to scheduled academic session	001
campus_id	VARCHAR	20	No	Campus identifier	Campus-01
lecture_datetime	DATETIME	–	No	Scheduled lecture start	2026-02-03 09:00:00
entry_time	DATETIME	–	Yes	Biometric entry scan time	2026-02-03 09:01:30
exit_time	DATETIME	–	Yes	Biometric exit scan time	2026-02-03 10:25:45
status	ENUM	–	No	Present, Late, Absent, Excused	Present

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capture_method	ENUM	–	No	Types of capture methods ("Biometric" or "Manual")	Biometric
leave_id	INT	11	Yes	FK to LeaveRequest (if Excused)	7001

Table C.4 – AttendanceRecord Data Dictionary

Figure E-2c: SRS Appendix C – AttendanceRecord Data Dictionary (Table C.4) showing full table.

status	ENUM	–	No	Present, Late, Absent, Excused	Present
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Figure E-2d: SRS Appendix C – AttendanceRecord status ENUM defined as "Present, Late, Absent, Excused" (close-up).

E.3 QC-FND-003: Contradiction in Report Generation Time Specification

- The administrator generates an attendance report, which the system returns within 10-60 seconds depending on data volume.

Figure E-3a: SRS Section 4.6.2 – report returns “within 10–60 seconds depending on data volume” without distinguishing query types.

- **NFR-PERF-03:** The system shall generate attendance reports within 10 seconds for single class queries and within 60 seconds for institution-wide reports.
 - Rationale: Administrators need timely reporting to manage compliance actions on five campuses. (TR-1-BE-01b)

Figure E-3b: SRS Section 5.1 – NFR-PERF-03 specifies 10 seconds for single class queries and 60 seconds for institution-wide reports.

E.4 QC-FND-004: Omission of Biometric Enrolment Process

3.2 Hardware Interfaces

The Biometric Student Attendance System interfaces directly with the biometric fingerprint scanning devices already procured and deployed by BioSec across all five campuses.

Hardware Technical Requirements

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- **FR-HW-01 (from BOR-0-03):** The system shall receive biometric scan data from fingerprint devices at all five BioSec campuses, each identified by a unique device_id registered in the system prior to deployment.
- **FR-HW-02 (from TR-1-BE-04):** Scan processing time from physical fingerprint contact to attendance outcome display shall not exceed 2 seconds per attempt, supporting a throughput of 50 students per device per 10 minutes during peak periods.
- **FR-HW-03 (from TR-1-BE-01c):** The system shall support up to 200 concurrent biometric scans across all campus devices simultaneously without exceeding the 2-second processing threshold or degrading backend API response times.
- **FR-HW-04 (from BOR-1-ATT-01):** After 3 consecutive failed scan attempts, the system shall initiate a supervised manual override flow and generate a flagged attendance record for administrator review.
- **FR-HW-05 (from BOR-0-03):** If a biometric scanner device becomes unavailable or fails during a session, students who were not scanned shall not be automatically marked absent. The device failure shall be logged with the device_id and timestamp for administrator action.

Figure E-4a: SRS Section 3.2 – Hardware Interface Requirements define biometric scanning behaviour (FR-HW-01 to FR-HW-05) but contain no enrolment process.

- **FR-ATT-03:** Following a successful biometric scan or an approved manual override, the system shall create an AttendanceRecord entry for each student, storing the student ID, session ID, campus, timestamp, and capture method. (BOR-1-ATT-01, BOR-1-ATT-02, UC-15)

Figure E-4b: SRS Section 4.3.3 – FR-ATT-03 creates AttendanceRecord entries assuming students are already enrolled, but no enrolment requirement exists.

E.5 QC-FND-005: Omission of Password Reset and Account Recovery

- **NFR-SEC-04:** The system shall lock all user accounts after a configurable number of failed login attempts, with a configurable lockout duration. All lockout events are logged with timestamp and account identifier. (TR-1-CPR-01)

Figure E-5a: SRS Section 5.3 – NFR-SEC-04 specifies account lockout after configurable failed login attempts.

If credentials are invalid, the system will display an error message and prompt re-entry. After five unsuccessful attempts, the account is locked and the user is notified.

Figure E-5b: SRS Section 4.2.2 – confirms lockout after five unsuccessful attempts, but no recovery mechanism is defined.

E.6 QC-FND-006: Omission of Offline Limit Exceeded Handling

- The system must support offline attendance viewing and leave request drafting for up to 48 hours, with a 50MB local storage limit.

Figure E-6a: SRS Section 2.5 – offline mode supports up to 48 hours with 50MB local storage limit.

- **NFR-SEC-09:** The system shall encrypt all locally cached data on mobile devices using AES-256. Data cached for offline use shall be automatically purged after 48 hours. (TR-1-CL-01f)

Figure E-6b: SRS Section 5.3 – NFR-SEC-09 confirms auto-purge after 48 hours, but no limit-exceeded or sync conflict behaviour is defined.

E.7 QC-FND-007: Omission of Session Creation and Management Requirements

- **BR-02:** Attendance capture only permitted during officially scheduled academic sessions. Scans outside session windows rejected. (BOR-1-ATT-01)

Figure E-7a: SRS Section 5.5 – BR-02 states attendance capture is only permitted during “officially scheduled academic sessions”.

- **FR-ATT-03:** Following a successful biometric scan or an approved manual override, the system shall create an AttendanceRecord entry for each student, storing the student ID, session ID, campus, timestamp, and capture method. (BOR-1-ATT-01, BOR-1-ATT-02, UC-15)

Figure E-7b: SRS Section 4.3.3 – FR-ATT-03 references session ID, but no requirement defines how sessions are created or managed.

E.8 QC-FND-008: Ambiguous Use of “Periodic Synchronisation”

- Attendance data access: role-scoped mobile access for parents (child only) and teachers (assigned classes only), with periodic synchronisation.

Figure E-8a: SRS Section 2.2 – Product Functions describes “periodic synchronisation” without specifying a frequency.

- **FR-COM-01 (from TR-1-CL-01c, BOR-1-MOB-02c):** Mobile applications shall synchronise attendance data with the backend every 5 minutes when connected, with a maximum data staleness of 30 minutes.

Figure E-8b: SRS Section 3.4 – FR-COM-01 specifies synchronisation “every 5 minutes when connected”.

E.9 QC-FND-009: Grammatical Error in QA-USAB-01

- **QA-USAB-01:** The system shall allow parent and teacher users must be able to complete primary tasks (view attendance, submit leave requests) within three interactions from the home screen, without referring to the user manual. The interface must accommodate users with varying technical literacy.
 - Verification: Entails conducting usability tests with representative parent and teacher participants before deployment.

Figure E-9: SRS Section 5.4.1 – QA-USAB-01 contains the double verb construction “shall allow ... must be able to”.

E.10 QC-FND-010: Capitalisation Inconsistency in UR-WEB-08 and UR-WEB-09

- **UR-WEB-07:** The portal shall allow administrators to configure the attendance compliance threshold percentage via the settings page.
- **UR-WEB-08:** the portal shall display an activity log for administrators to view system notifications
- **UR-WEB-09:** the portal shall include a 2-Factor Authentication screen during the authentication process.
- **UR-WEB-10:** The portal shall include a bulk selection of students via checkboxes in the student attendance records to implement manual attendance override.

Figure E-10: SRS Section 3.1 – UR-WEB-08 and UR-WEB-09 begin with lowercase “the portal shall”, inconsistent with all other requirements.

E.11 QC-FND-011: Unsigned SRS Sign-off Section

Designated Approvers – Vendor (Nevon Solutions)			
Role	Name	Signature	Date
Technical Lead	James Koh		
System Architect Lead	Rachel Lee		

Designated Approvers – Customer (BioSec)			
Role	Name	Signature	Date
Project Manager	Marcus Tan Wei Liang		
Head of Operations	Daniel Rodriguez		

Figure E-11: SRS Section 9 – Designated Approvers tables with entirely blank Signature and Date fields for all four approvers.

E.12 QC-FND-012: RBAC Not Defined in Glossary

Appendix A Glossary	
Term	Definition
AES-256	Advanced Encryption Standard with 256-bit key length for biometric data encryption at rest.
APNS	Apple Push Notification Service for iOS push notifications.
API	Application Programming Interface for system integrations.
AR	Analysing Report (NS-AR-2026-001) by BioSec.
Azure SQL	Microsoft Azure cloud-based relational database service.
Biometric Template	Digital representation of fingerprint stored for authentication comparison.
BOR	Business Operations Requirement from the PS.
CAPEX	Capital Expenditure; one-time project investment costs.
DFD	Data Flow Diagram showing system data flows and external entities.
FAR	False Acceptance Rate; target <0.01% per TC-02 revision.
Firebase	Google Firebase Cloud Messaging for Android push notifications.
FR	Functional Requirement specifying system behaviour.
FRR	False Rejection Rate; target <1%.
IEEE 830	IEEE Std 830-1998 for SRS structure (basis of this template).
iRTM	Inceptive Requirements Traceability Matrix; PS to SRS chain of custody.
ISO/IEC 19794-2	International standard for fingerprint minutiae data interchange.
MC	Medical Certificate for illness/injury documentation.
MCS	Medical Certificate System; external integration for MC verification.
NFR	Non-Functional Requirement specifying quality or constraints.
OAuth 2.0	Authorization framework for secure delegated API access.
PDPA	Personal Data Protection Act 2012 (Singapore).
PS	Project Specification (BS-PS-2026-001) by BioSec.
RPO	Recovery Point Objective; maximum acceptable data loss duration.
RTO	Recovery Time Objective; maximum acceptable downtime duration.
SLA	Service Level Agreement between parties.

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SMTP	Simple Mail Transfer Protocol for email notifications.
SRS	Software Requirements Specification (this document, NS-SRS-2026-001).
TLS	Transport Layer Security for encrypted communications.
TR	Technical Requirement from the PS.

Table A.1 – Glossary of Terms

Figure E-12: SRS Appendix A – Glossary defines 27 terms but does not include RBAC, despite its extensive use throughout the SRS.

References

- BioSec Educational Institution, *Project Specification: Biometric Student Attendance System* (BS-PS-2026-001, Version 1.0), ICT2113, 2026.
- Nevon Solutions, *Software Requirements Specification (SRS): Biometric Student Attendance System* (NS-SRS-2026-001, Version 0.8), 3 March 2026.
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